

DengAI — Predicting Disease Spread

Data, Metrics, Models and Preparation

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Contents

Part 1 — The data and the problem	1
Part 2 — Metrics, models and preparation	6
Metrics	6
Models	7
Notebook Preparation	9
Part 3 — Additional EDA	11

Part 1 — The data and the problem

Goal: estimate reported dengue cases each week in San Juan, and Iquitos using environmental measurements.

Load the data and take a first look

((1456, 24), (1456, 4), (416, 24))

1,456 historical weeks, 416 future weeks, and 24 input columns. The labels contain the city-week identifiers and the outcome.

Column	Row 1	Row 2
city	sj	sj
year	1990	1990
weekofyear	18	19
week_start_date	1990-04-30	1990-05-07
ndvi_ne	0.1226	0.1699
ndvi_nw	0.103725	0.142175
ndvi_se	0.198483	0.162357
ndvi_sw	0.177617	0.155486
precipitation_amt_mm	12.42	22.82
reanalysis_air_temp_k	297.573	298.211
reanalysis_avg_temp_k	297.743	298.443
reanalysis_dew_point_temp_k	292.414	293.951
reanalysis_max_air_temp_k	299.8	300.9
reanalysis_min_air_temp_k	295.9	296.4
reanalysis_precip_amt_kg_per_m2	32	17.94
reanalysis_relative_humidity_percent	73.3657	77.3686
reanalysis_sat_precip_amt_mm	12.42	22.82
reanalysis_specific_humidity_g_per_kg	14.0129	15.3729
reanalysis_tdtr_k	2.62857	2.37143
station_avg_temp_c	25.4429	26.7143

Column	Row 1	Row 2
station_diur_temp_rng_c	6.9	6.37143
station_max_temp_c	29.4	31.7
station_min_temp_c	20	22.2
station_precip_mm	16	8.6

The environmental features come from four sources 1. Weather stations (5 features)

Ground stations measure local average, minimum and maximum temperature, daily temperature range, and rainfall. For example, `station_avg_temp_c` describes average station temperature.

2. Satellite rainfall (1 feature)

`precipitation_amt_mm` measures total rainfall in **mm** over a $0.25^\circ \times 0.25^\circ$ grid cell. It provides an area-based rainfall measurement alongside the station observations.

3. Climate reanalysis (10 features)

Reanalysis combines observations with a weather model to estimate environmental conditions on a consistent grid. The `reanalysis_` columns describe temperature, dew point, humidity and precipitation at $0.5^\circ \times 0.5^\circ$ resolution. Temperature levels are in **kelvin**, relative humidity in %, and specific humidity in **g/kg**. These describe both heat and moisture conditions.

4. Satellite vegetation (4 features)

The **Normalized Difference Vegetation Index (NDVI)** describes vegetation around each city using $0.5^\circ \times 0.5^\circ$ pixels. The columns `ndvi_ne`, `ndvi_nw`, `ndvi_se` and `ndvi_sw` refer to pixels north-east, north-west, south-east and south-west of the city centre.

Group	Column names
Location and time	<code>city</code> , <code>year</code> , <code>weekofyear</code> , <code>week_start_date</code>
Station temperature	<code>station_avg_temp_c</code> , <code>station_min_temp_c</code> , <code>station_max_temp_c</code> , <code>station_diur_temp_rng_c</code>
Station and satellite rainfall	<code>station_precip_mm</code> , <code>precipitation_amt_mm</code>
Reanalysis temperature	<code>reanalysis_air_temp_k</code> , <code>reanalysis_avg_temp_k</code> , <code>reanalysis_min_air_temp_k</code> , <code>reanalysis_max_air_temp_k</code> , <code>reanalysis_dew_point_temp_k</code> , <code>reanalysis_tdtr_k</code>
Reanalysis humidity	<code>reanalysis_relative_humidity_percent</code> , <code>reanalysis_specific_humidity_g_per_kg</code>
Reanalysis precipitation	<code>reanalysis_precip_amt_kg_per_m2</code> , <code>reanalysis_sat_precip_amt_mm</code>
Vegetation	<code>ndvi_ne</code> , <code>ndvi_nw</code> , <code>ndvi_se</code> , <code>ndvi_sw</code>

Source: [DrivenData feature definitions](#).

	city	year	weekofyear	total_cases
0	sj	1990	18	4
1	sj	1990	19	5
2	sj	1990	20	4
3	sj	1990	21	3
4	sj	1990	22	6

total_cases is the value to predict: a weekly count of reported cases. For example, the first San Juan week has **4 cases**.

Check the cities and their timelines

San Juan (sj) and Iquitos (iq) have different amounts of history.

city	count	min	max
iq	520	2000-07-01	2010-06-25
sj	936	1990-04-30	2008-04-22

city	count	min	max
iq	156	2010-07-02	2013-06-25
sj	260	2008-04-29	2013-04-23

Each city's test period follows its training period: approximately **five years** for San Juan and **three years** for Iquitos. We therefore validate on later weeks.

Next, we examine differences between cities.

The cities have different case levels and seasons

city	median	max
Iquitos	5	116
San Juan	19	461

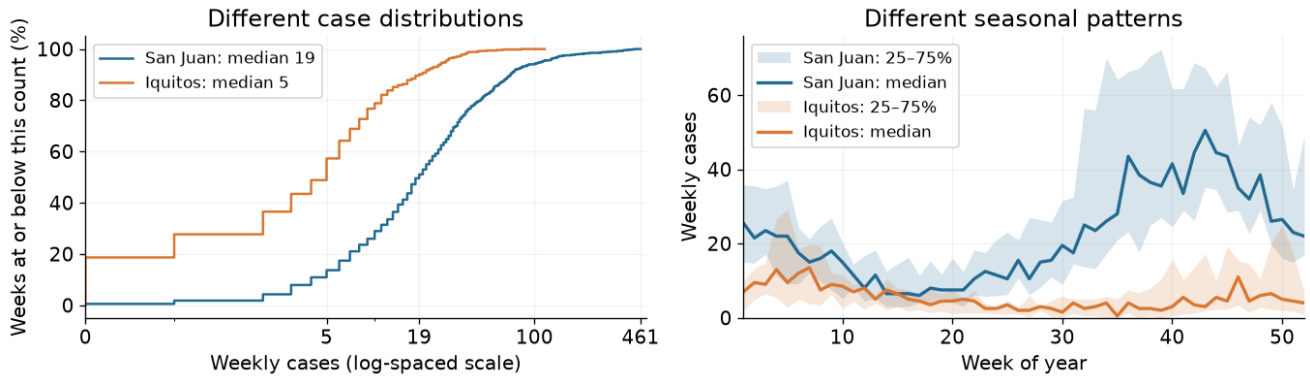


Figure 1: Case levels and seasonal timing differ between the two cities

The charts show differences in both the distribution of cases and the time of year when cases are higher.

Decision: perform EDA separately, then prepare and fit a model for each city.

Missing Data

Missing measurements across all inputs:

```
ndvi_ne          194
ndvi_nw          52
station_diur_temp_rng_c  43
station_avg_temp_c  43
station_precip_mm  22
dtype: int64
```

Vegetation (NDVI) have the most missing values

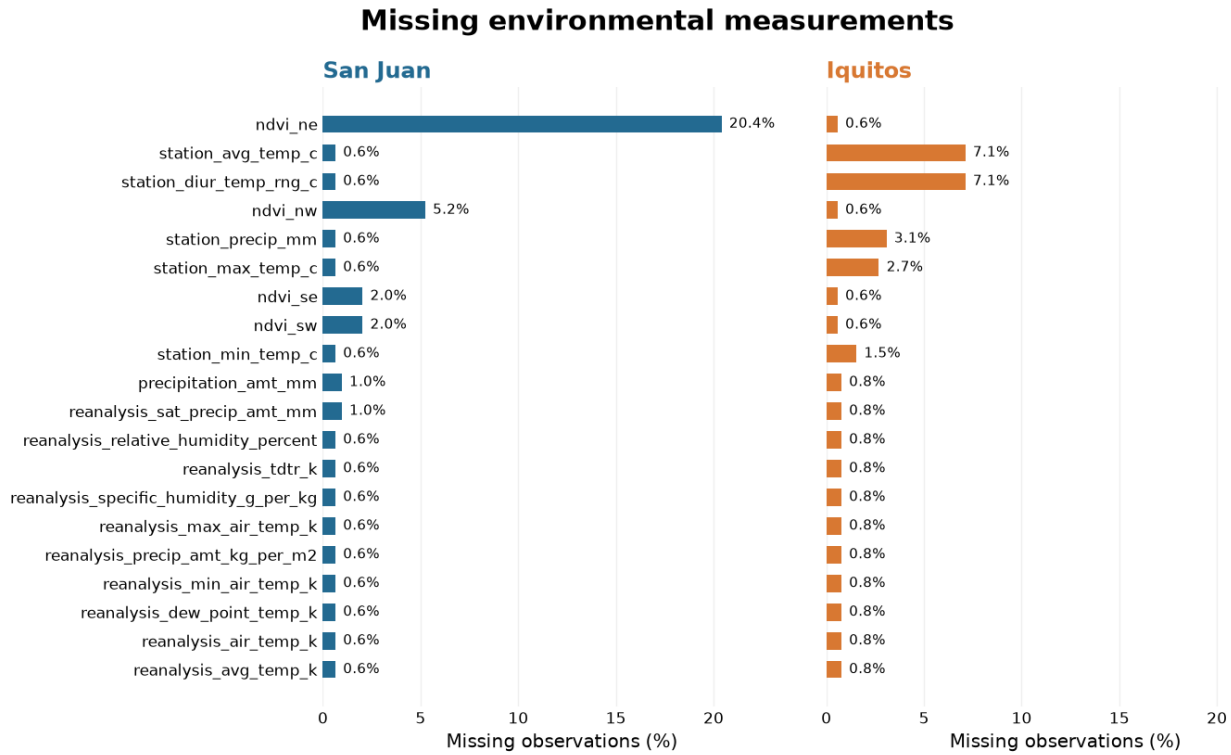


Figure 2: Missing measurements by column and city

Compare the missing percentage and longest consecutive gap within each city:

City	Vegetation feature	Missing (%)	Longest gap (observations)
San Juan	ndvi_ne	20.4	15
San Juan	ndvi_nw	5.2	15
San Juan	ndvi_se	2	14
San Juan	ndvi_sw	2	14
Iquitos	ndvi_ne	0.6	1
Iquitos	ndvi_nw	0.6	1
Iquitos	ndvi_se	0.6	1
Iquitos	ndvi_sw	0.6	1

How we fill missing measurements:

- **San Juan vegetation:** use the typical value (median) for the same week of the year, learned from the training data. If unavailable, use the overall training median for that measurement.
- **Iquitos vegetation:** estimate missing values between the nearest available measurements before and after the gap (linear interpolation).
- **Other weather measurements in both cities:** use the same interpolation approach for gaps between known values.

Why treat San Juan vegetation differently? - Its long gaps would otherwise repeat an old value for many weeks if we used forward-fill.

Weather history

Why look at earlier weather?

- **The effect may take time:** weather conditions today may be linked to dengue cases several weeks later. Earlier measurements help the model capture this delay.
- **Good conditions may build up:** our hypothesis is that several weeks of suitable warmth and moisture allow mosquito populations to grow more than one favourable week alone.
- **Duration matters:** rolling averages of temperature and humidity, and rainfall totals, describe whether conditions stayed favourable over several weeks.

What did we find and use?

- **In San Juan:** temperature and specific humidity have stronger positive associations with current cases around **8–10 observations earlier** than in the same week. The pattern differs in Iquitos.
- **Weather lags:** include measurements from **1, 2, 4, 8, 10, 20 and 26 observations earlier**.
- **Rolling summaries:** describe weather over the previous **4–26 observations** to capture sustained conditions.
- **Benchmark motivation:** DrivenData also suggests exploring earlier weather to address a mismatch in the timing of predicted cases. [Benchmark discussion](#)

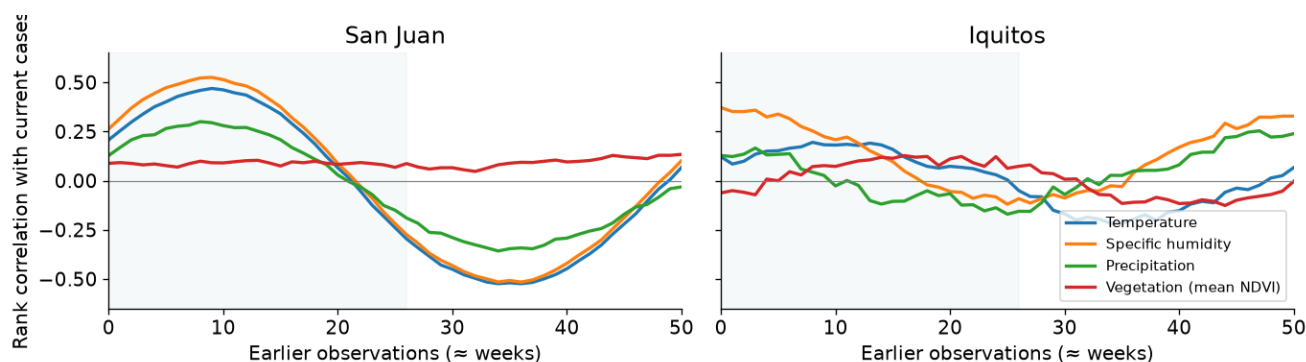


Figure 3: Evidence that weather acts on cases with a lag

Cross-validation

- **Keep time order:** train on earlier weeks and validate on later weeks.
- **Repeat five times per city:** predict the next **52 observations** (about one year), using more training history each round.
- **Keep a final check:** set aside the last **52 observations** as a separate holdout.
- **Choose settings using CV MAE:** lower average error is better. Then report the holdout error.
- **Why:** a random split would mix earlier and later periods. Chronological CV asks a question closer to the competition task.

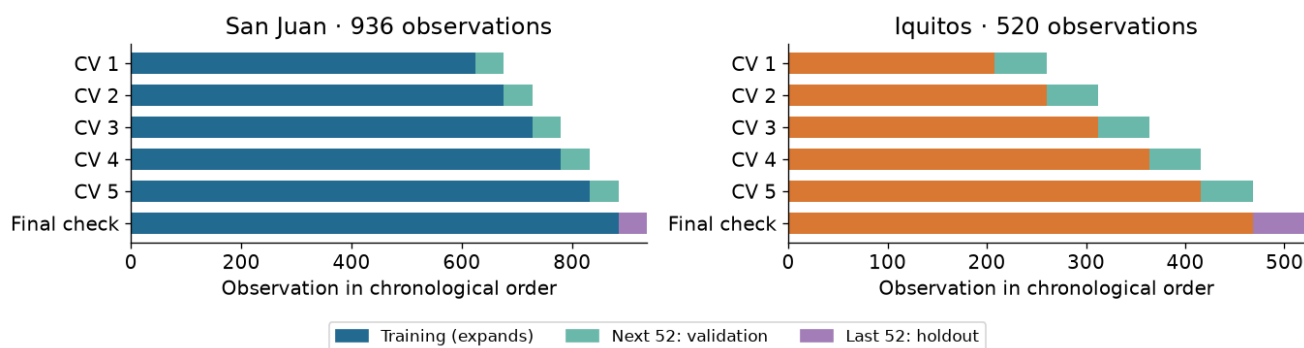


Figure 4: Chronological cross-validation timeline

Pipelines

- **One process:** fill gaps, add weather history, prepare inputs and train the model.
- **Avoid data leakage:** learn filling rules, scaling and feature selection from training data only.
- **Fair validation:** keep validation case counts out of training.
- **Consistent predictions:** reuse the same fitted preparation for new data.

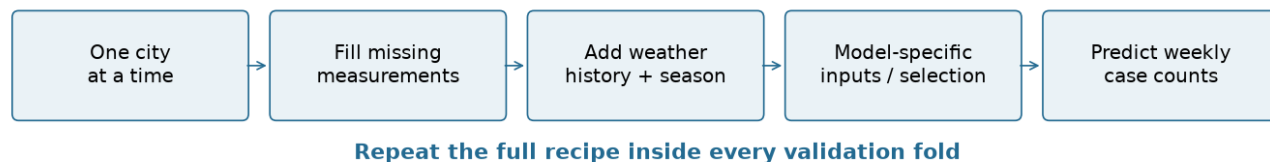


Figure 5: The modelling pipeline

Part 2 — Metrics, models and preparation

Metrics

The competition metric is MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

The average number of cases we are off by, per week.

- Reported over all **416 test weeks** (260 San Juan + 156 Iquitos)
- Lower is better; a perfect model scores 0
- Both cities are pooled into one number, so **San Juan dominates** — it has 63% of the test weeks *and* roughly four times the case counts

Why MAE and not RMSE

The target is extremely skewed:

	San Juan	Iquitos
mean cases/week	34.2	7.6
median	19	5
max	461	116
skewness	4.48	4.00

In both cities the **worst 10% of weeks hold 43% of all cases**.

Why MAE and not RMSE (cont.)

RMSE squares the errors, so a single epidemic week with 461 cases contributes as much as **~500 ordinary weeks** being off by one.

- Under RMSE the model is dragged into fitting a handful of outbreaks
- MAE weights every week equally
- Statistically: **MAE is minimised by the conditional median, RMSE by the mean** — and for a skewed count distribution the median is the more robust target

This also explains a behaviour we see later: good models here are *conservative*.

Two different MAEs — do not confuse them

	Cross-validation MAE	Holdout MAE
Computed on	5 expanding-window folds	one untouched 52-week block
Used for	choosing hyperparameters	comparing final models
Seen during tuning?	yes — the grid optimises it	no

CV MAE is the *tuning signal*. Holdout MAE is the *honest estimate*. They are never equal, and a large gap between them is itself information: it means the model is overfitting the folds.

Why validation must be chronological

Weekly case counts are **autocorrelated at 0.97** from one week to the next.

A random train/test split would therefore put week t in training and week $t + 1$ in test — the model effectively memorises the answer and the score becomes meaningless.

We use `TimeSeriesSplit(n_splits=5, test_size=52)`: always **train on the past, validate on the future**.

The fold structure

City	Fold	Train ends	Validation
San Juan	1	2002-04-23	2002-04-30 → 2003-04-23
San Juan	3	2004-04-22	2004-04-29 → 2005-04-23
San Juan	5	2006-04-23	2006-04-30 → 2007-04-23
Iquitos	1	2004-06-24	2004-07-01 → 2005-06-25
Iquitos	5	2008-06-24	2008-07-01 → 2009-06-25

Expanding window: each fold trains on everything before it and validates on the next 52 weeks. The final 52 weeks of each city are held back entirely.

The constraint is enforced, not just intended

`SARIMAXRegressor.predict()` **raises an exception** if asked to predict a block that is not strictly after its training data:

```
if pd.to_datetime(X["week_start_date"]).iloc[0] <= self.train_end_:
    raise ValueError("SARIMAX needs a future block "
                     "after its training data.")
```

This makes temporal leakage a crash rather than a silently optimistic score. No shuffling, no `cross_val_predict`, anywhere in the project.

Models

The pipeline has four branches

Different model families need different preprocessing, so `make_city_pipeline` builds one of four shapes:

Branch	Steps	Models
<code>tree</code>	impute → history → cyclical → select → model	CatBoost, RandomForest
<code>dense</code>	tree steps + impute + scale	Ridge, Lasso
<code>time_series</code>	impute → history → date + scaled weather	Prophet, SARIMAX

Branch	Steps	Models
<code>seasonal</code>	calendar columns only	Seasonal median, $\text{Shape} \times \text{level}$

Why `dense` exists

The `tree` branch leaves **NaNs** in the lag/rolling warm-up — the first 26 weeks have no 26-week history.

- Tree models (sklearn ≥ 1.4) **handle NaN natively**
- RidgeCV raises `ValueError: Input X contains NaN`

So linear models get two extra steps: `SimpleImputer(median)` then `StandardScaler`. Scaling matters for them too, since Ridge and Lasso penalise coefficients and are not scale-invariant.

Why `seasonal` exists

The tree branch **drops** `city`, `week_start_date` and `weekofyear` before the model.

But the two calendar baselines need exactly those columns. Rather than weaken the main branch, they get their own: a `ColumnTransformer` passing through `year` and `weekofyear`, nothing else.

These models **never see the weather at all** — which is the point: they are the baseline any weather-driven model should have to beat.

The eight models

Model	Family	Core assumption
CatBoost	gradient boosting	non-linear interactions of lagged weather
RandomForest	bagged trees	same, without boosting
Ridge	linear + L2	smooth linear response, all features shrunk
Lasso	linear + L1	linear, but most features are irrelevant
Prophet	additive time series	trend + annual seasonality + regressors
SARIMAX	ARIMA + exog	autocorrelated errors + Fourier seasonality
Seasonal median	calendar only	this week behaves like this week in past years
$\text{Shape} \times \text{level}$	calendar only	seasonal <i>shape</i> $\times$ typical annual <i>total</i>

Results — holdout MAE

Model	San Juan	Iquitos
CatBoost	15.02	6.27
RandomForest	15.16	8.39
Shape $\times$ level	20.72	3.39
Lasso	21.16	4.50
Prophet	21.78	7.16
Seasonal median	23.49	4.41
Ridge	24.46	5.48
SARIMAX	24.56	4.61

Different winners per city — so the final submission uses **CatBoost** for **San Juan**, **SARIMAX** for **Iquitos**.

The most interesting result

The best Iquitos model never looks at the weather.

Shape × level scores **3.39**, beating SARIMAX (4.61) and CatBoost (6.27).

It works in two parts:

1. **Shape** — the seasonal profile, normalised to sum to 1
2. **Level** — the median annual total of complete training years

$$\hat{y}_{\text{week}} = \text{shape}(\text{week}) \times \text{level}$$

Why that result makes sense

The decomposition matches what the EDA found:

- Climate reliably predicts **when** cases rise within a year
- It explains very little about **how many**

Epidemic magnitude is driven by which dengue **serotype** is circulating and by population immunity — neither is in this dataset.

So a model that predicts the *shape* from the calendar and refuses to guess the *level* from weather is not naive. It is **honest about what the data supports**.

Notebook Preparation

Missing values differ by city

	San Juan	Iquitos
ndvi_ne missing	191 / 936 (20.4%)	3 / 520 (0.6%)
ndvi_nw	49	3

San Juan is coastal: the north-east NDVI pixel often falls over the **Atlantic**, where the vegetation index is undefined, and cloud over water blocks the satellite.

One imputation strategy for both cities would be wrong.

DengueImputer — two strategies

Columns	Method	Why
NDVI (San Juan only)	week-of-year training median , then overall median	20% missing in long runs — interpolation would invent a trend
Everything else	linear interpolation, <code>limit_area="inside"</code>	weather is smooth in time; a gap is a sensor outage

`limit_area="inside"` matters: it refuses to extrapolate past the first and last real observation, so we never fabricate data at the edges.

Feature engineering — 20 columns become 124

From **8 weather variables**:

- **Lags** at 1, 2, 4, 8, 10, 20, 26 weeks → $8 \times 7 = 56$ columns
- **Rolling means** over 4, 8, 10, 12, 16, 26 weeks → $6 \times 6 = 36$
- **Rolling sums** (rainfall accumulates, it does not average) → $2 \times 6 = 12$

104 engineered columns on top of the 20 originals.

Lags matter because the biology is delayed: weather → mosquito breeding → infection → symptoms → diagnosis takes weeks.

The leakage trap in rolling features

A rolling mean computed on the **whole dataset** leaks the future into the past.

WeatherHistoryTransformer solves this:

```
def fit(self, X, y=None):
    self.history_ = X.tail(max_history).copy()

def transform(self, X):
    history = self.history_.loc[
        self.history_["week_start_date"] < X["week_start_date"].iloc[0]]
    result = pd.concat([history, X])
    ...
    return result.iloc[-len(X):]
```

Why that design is right

fit memorises only the **tail of the training fold**. transform prepends only rows **strictly earlier** than the incoming block.

So a validation block gets lag features computed from real past data — not NaN, and not future data.

Every window is also `.shift(1)` before rolling, so **the current week is never included in its own average**.

Why a Pipeline and not inline code

Every preprocessing step is **fitted**: medians, scalers, the feature selector, the history tail.

If preprocessing happens before the split, those statistics are computed on data the model will later be tested on — a subtle leak that inflates every score.

Inside a Pipeline, GridSearchCV refits **the whole chain** on each fold's training part only. Correctness becomes structural rather than something to remember.

The refactor — notebook to package

All machinery moved out of the notebook into an importable package:

```
DengAI/
  transformers.py  DengueImputer, WeatherHistoryTransformer
  selectors.py     CatBoostFeatureSelector
  models/         Prophet, SARIMAX, seasonal baselines
  pipelines.py    make_city_pipeline --- the four branches
  evaluate.py     CV, grid search, holdout, submission
configs/
  features.yaml   columns, lags, windows, imputation
  models.yaml     params, grids, branch per model
```

Why configuration lives in YAML

One rule keeps it simple:

Every mapping in the YAML is exactly the `__init__` keyword arguments of the object it configures.

```
WeatherHistoryTransformer(**cfg.features.weather_history)
```

No schema layer, no validation framework. A typo becomes a `TypeError` when the pipeline is built — which is the failure you want, loud and immediate.

Changing the lag grid or adding a model is a config edit, not a code change.

Verification — we proved the move was safe

Before deleting anything from the notebook, the extracted classes were run **side by side** with the originals on real data:

```
sj  DengueImputer          IDENTICAL  (936, 24)
sj  WeatherHistoryTransformer IDENTICAL  (52, 128)
iq  DengueImputer          IDENTICAL  (520, 24)
iq  WeatherHistoryTransformer IDENTICAL  (52, 128)
```

And the package reproduces the original scores: **SARIMAX San Juan holdout 24.564 vs 24.564 recorded** — **exact**, Iquitos CV 6.982 exact, identical fold boundaries.

Three real bugs found along the way

- 1. The notebook could not run in parallel.** CatBoost's `select_features` ignores `allow_writing_files=False` and does a bare `mkdir("catboost_info")`. With `n_jobs=2`, two workers race and one dies with `FileExistsError`. Fixed with a per-fit `train_dir`.
- 2. KeyError: 'catboost'** in the holdout cells — they refit from a dict only populated for models loaded from disk. They now read predictions already computed.
- 3. The submission model choice was declared twice, inconsistently.** Now stated once, in `models.yaml`.

Summary of part 2

Metrics — MAE because the target is skewed (skewness 4.5); CV MAE tunes, holdout MAE judges; validation is chronological because adjacent weeks correlate at 0.97.

Models — eight models across four pipeline branches, because tree, linear, time-series and calendar families need different preprocessing. Winners differ by city.

Preparation — city-specific imputation, 104 leakage-safe engineered features, and a **Pipeline** so that correctness is structural. All of it extracted into a package with YAML configuration, verified byte-identical before the originals were removed.

Part 3 — Additional EDA

Setup

Weekly case counts per city:

City	count	mean	std	min	25%	50%	75%	max
Iquitos	520	7.57	10.77	0	1	5	9	116
San Juan	936	34.18	51.38	0	9	19	37	461

Weekly cases

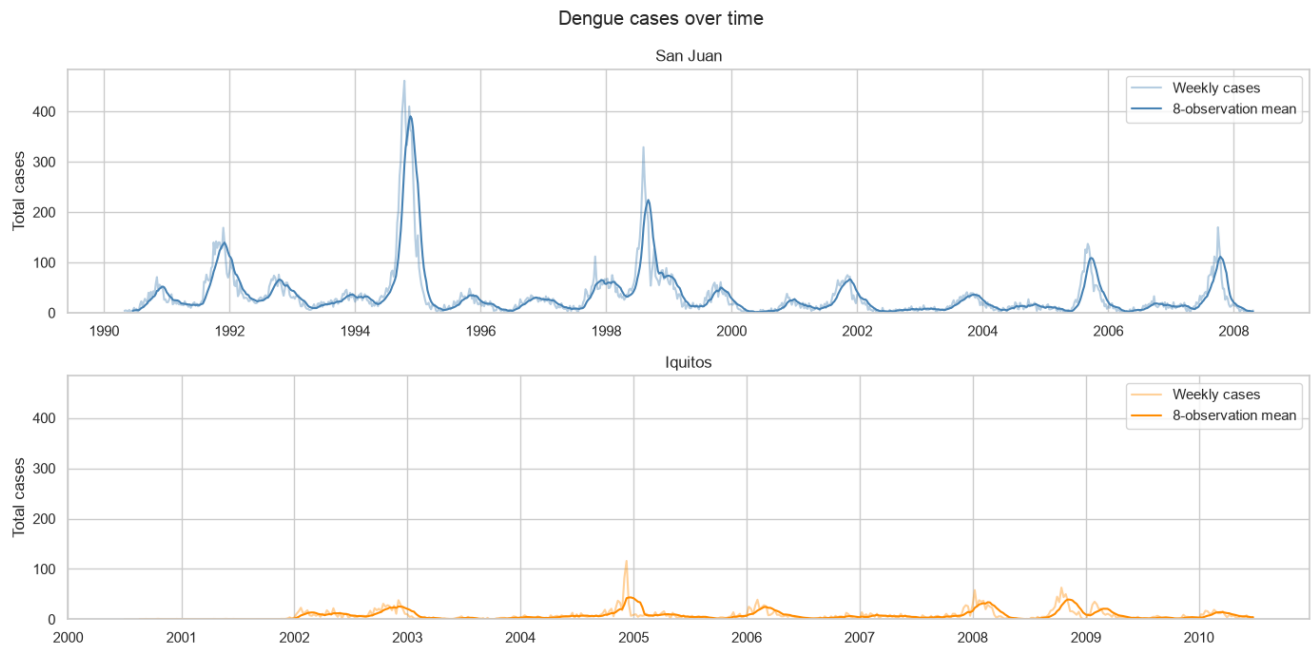


Figure 6: Weekly dengue cases over time, by city

Seasonality

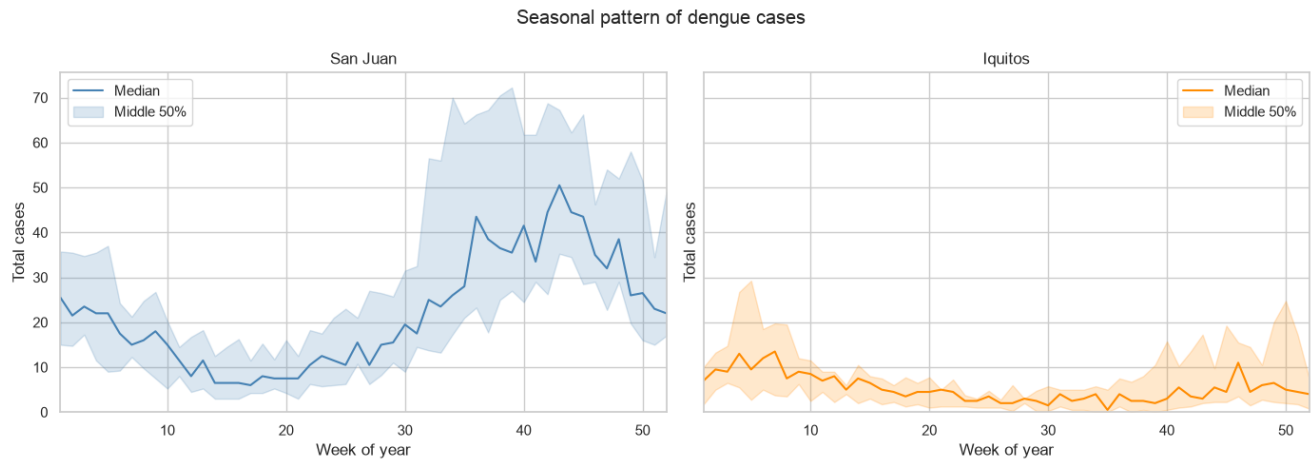


Figure 7: Seasonal profile: median cases by week of year

San Juan tends to have higher counts later in the year, while Iquitos has higher medians near the beginning. This supports exploring seasonal features separately for the cities.

Weather and cases

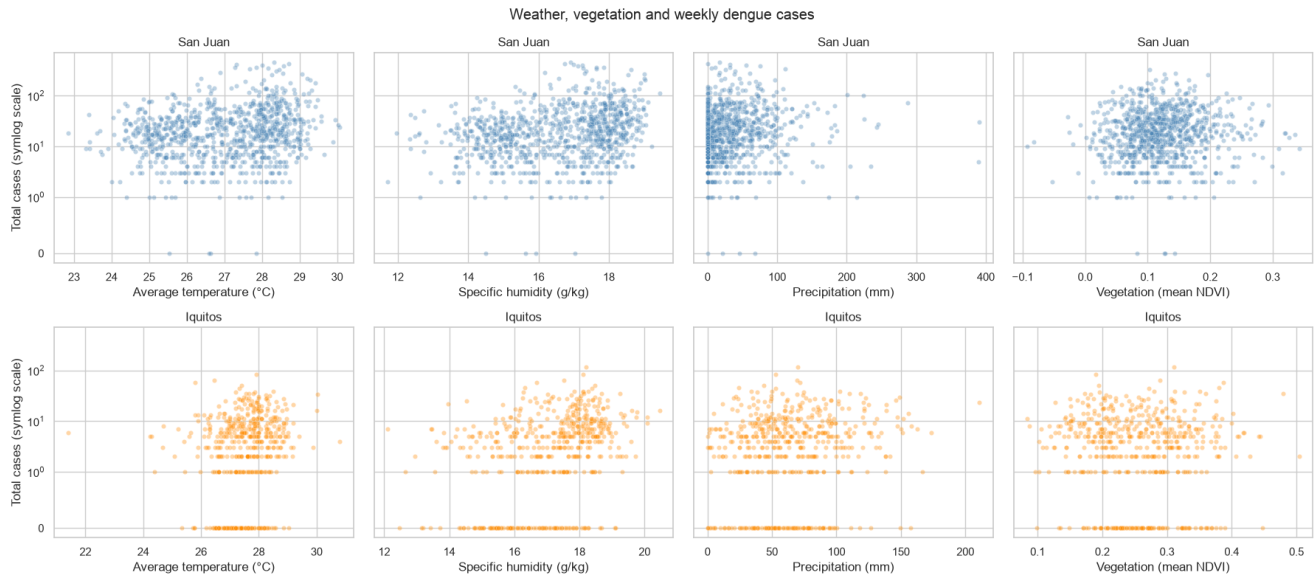


Figure 8: Weather and vegetation against case counts

Missing values

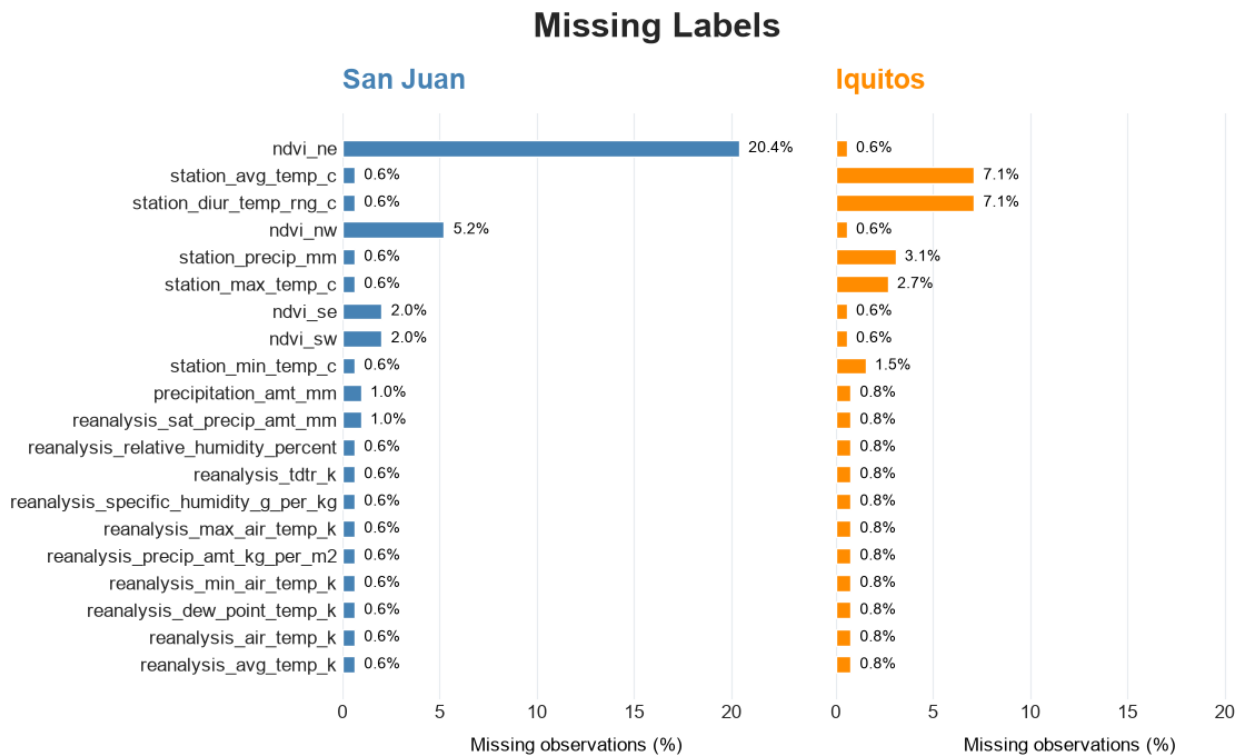


Figure 9: Missing values by column and city

Weather lags

Compare earlier weather and vegetation with current cases using Spearman correlation. A lag of 4 means four rows earlier, approximately four weeks.

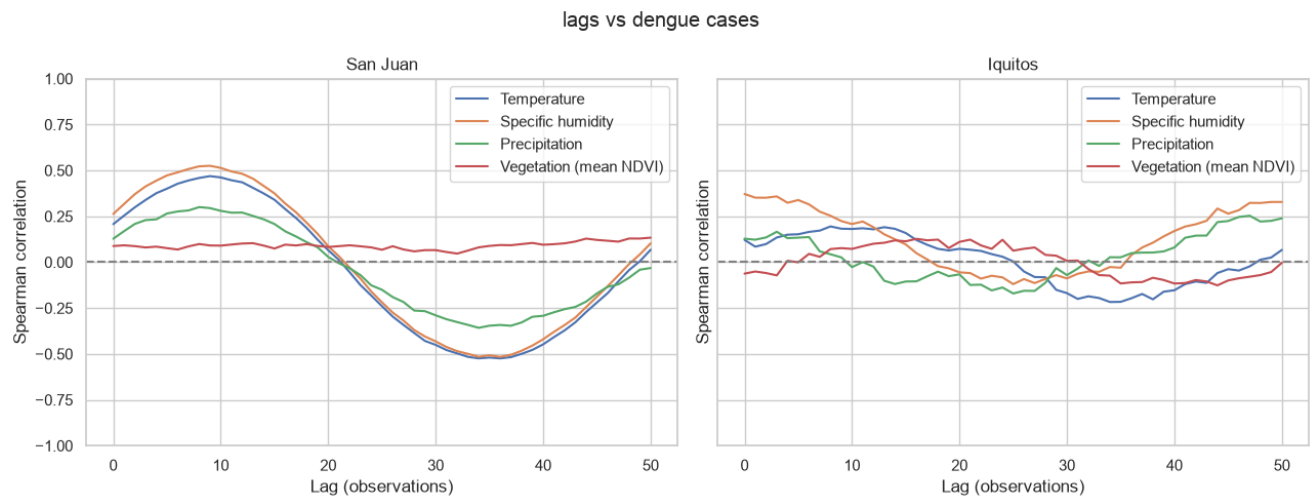


Figure 10: Spearman correlation of lagged weather with current cases